

Shuaicheng Zhang

PHD CANDIDATE · COMPUTER SCIENCE DEPARTMENT OF VIRGINIA TECH

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Research Interest

My research focuses on **developing foundation models and graph-augmented AI systems** that bridge **structured knowledge** with **large language models** for enhanced reasoning and decision-making capabilities. I specialize in **open-world graph learning**, **multimodal learning**, and **LLM-driven applications**, with particular emphasis on integrating graph neural networks with language models to handle complex relational reasoning tasks. My work spans temporal knowledge graphs, graph-enhanced LLM architectures, and structured reasoning systems, with applications in financial intelligence, industrial AI, scientific discovery, and knowledge-augmented generation.

Education

Virginia Tech

PHD CANDIDATE COMPUTER SCIENCE

- Advisor: Dr. Dawei Zhou

Blacksburg, VA

August 2020 - present

MIT CSAIL

VISITING PHD STUDENT

- Advisor: Dr. Julian Shun

Cambridge, MA

May 2023 - Aug 2023

University of Maryland, College Park

MS TELECOMMUNICATIONS

- Advisor: Dr. Michael Dellomo

College Park, MD

August 2018 - May 2020

Virginia Tech

BS COMPUTER SCIENCE

- Minors in Mathematics
- Advisor: Dr. T.M. Murali

Blacksburg, VA

August 2013 - Dec 2017

Industrial Experience

Gray Systems Lab, Microsoft Research

RESEARCH INTERN

- Developed **WorkloadGPT**, a multimodal foundation model integrating GNNs and LLMs to jointly reason over structured knowledge (SQL queries and execution plans) and temporal telemetry data for intelligent database workload identification and optimization across Azure services
- Built data pipelines processing millions of SQL workloads, achieving SOTA performance on workload identification task

Redmond, WA, USA

Summer 2025

CSAIL, MIT-IBM Watson AI Lab

RESEARCH INTERN

- Developed **H²GB** benchmark for evaluating GNNs on heterogeneous and **H²G-former** for a modular graph transformer with capability of large scale graph learning, winning **KDD 2025 Best Paper Award**
- Designed **UnifiedGT**, a unified graph transformer framework, bridging effective and efficient approaches for scalable systems handling billions of edges, published in IEEE BigData 2024

Cambridge, MA, USA

Summer 2023

Natural Language Processing Team, Deloitte AI Center of Excellence

NLP RESEARCH INTERN

- Designed open relation extraction system based on GPT-3 for analyzing important events and relationships from audit files, helping auditors better identify critical patterns and connections in financial documents

New York, NY, USA

Summer 2021

Trust Team, Hundsun Technologies Inc.

SOFTWARE DEVELOPMENT ENGINEER INTERN

Hangzhou, China

Summer 2017

- Designed and implemented high-performance multithreaded PDF converter with concurrent processing workflows for financial document processing at scale

Preprints & Publications

Scientific Hypothesis Generation and Validation: Methods, Datasets, and Future Directions —Adithya Kulkarni, Fatimah Alotaibi, Xinyue Zeng, Longfeng Wu, Tong Zeng, Barry Menglong Yao, Minqian Liu, **Shuaicheng Zhang**, Lifu Huang, Dawei Zhou. In Arxiv, Under review

CAPTAIN: Conformal-Prediction-Based Multi-Source Time-Series Forecasting —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. Under review of **TMLR**'2025.

PDMBench: A Standardized and Multi-Perspective Benchmark for Predictive Maintenance on Multi-Sensor Industrial Time-Series Data —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. Under review of **ICLR**'2025.

HeroFilter: Adaptive Spectral Graph Filter for Varying Heterophilic Relations —**Shuaicheng Zhang**, Haohui Wang, Junhong Ling, Xiaojie Guo, Yada Zhu, Si Zhang, Dongqi Fu, Dawei Zhou. (*Equal Contribution) In proceedings of the (**NeurIPS**)'2025.

When Heterophily Meets Heterogeneity: New Graph Benchmarks and Effective Methods —Junhong Lin, Xiaojie Guo, **Shuaicheng Zhang**, Yada Zhu, Julian Shun. In Proceedings of the ACM **KDD**'2025. **Best Paper Award**

MentorPDM: Learning Data-Driven Curriculum for Multi-Modal Predictive Maintenance —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. In Proceedings of the ACM **KDD**'2025.

UnifiedGT: Towards a Universal Framework of Transformers in Large-Scale Graph Learning —Junhong Lin, Xiaojie Guo, **Shuaicheng Zhang**, Yada Zhu, Dawei Zhou, Julian Shun. In Proceedings of the **IEEE BigData** 2025. **Patent Pending**.

TGEditor: Task-Guided Graph Editing for Augmenting Temporal Financial Transaction Networks —**Shuaicheng Zhang**, Yada Zhu, Dawei Zhou. In Proceedings of the ACM International Conference on AI in Finance(**ICAIF**)'2023(oral). **Patent Pending, Deployed at Wellsfargo**.

Personalized Federated Learning under Mixture of Distributions —**Shuaicheng Zhang***, Yue Wu*, Yanchi Liu, Quanquan Gu, Dawei Zhou, Haifeng Chen, Wei Cheng. (*Equal Contribution) In Proceedings of International Conference on Machine Learning(**ICML**)'2023.

Extracting Temporal Event Relation with Syntax-guided Graph Transformer —**Shuaicheng Zhang**, Ning Qiang, Lifu Huang. In Findings of North American Chapter of the Association for Computational Linguistics(**NAACL**)'2022.

Awards & Honors

- 2025 **KDD Best Paper Award in Benchmark and Dataset Track**, KDD 2025
- 2025 **KDD Outstanding Reviewer (Top 10%)**, KDD 2025
- 2024 **CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
- 2023 **CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
- 2022 **CIKM Travel Award**, CIKM 2022
- 2021 **Virginia Tech Pratt Fellowship**, Virginia Tech Department of Computer Science

Academic Service

CONFERENCE REVIEWER

2026 **AAAI 2026, ICLR2026**, Reviewer, Program Committee
 2025 **KDD 2025, ICML 2025, ICLR 2025, Neurips 2025**, Reviewer, Program Committee
 2024 **KDD 2024, ICML 2024**, Reviewer, Program Committee
 2023 **KDD 2023, ICDM 2023, WWW 2023**, Reviewer
 2022 **SIGIR 2022, KDD 2022, NLPCC 2022, WWW 2023**, Reviewer
 2021 **SDM 2022, WWW 2022, CIKM 2022**, Subreviewer
 2021 **NLPCC 2021**, Reviewer

ORGANIZING COMMITTEE

2022 **The 1st Workshop on Trustworthy Learning on Graphs (TrustLOG) at The Conference on Information and Knowledge Management (CIKM)'2022**, Publicity Chair *Atlanta*

Skills

STATISTICS Advanced statistical analysis, inference, factor, regression analyses, hypothesis testing, hierarchical modeling

MACHINE LEARNING AND DEEP LEARNING TensorFlow, Keras, Pytorch, Scikit-Learn, XGBoost, Pandas, Numpy, Scipy, NLTK, NetworkX

PROGRAMMING Python, Java, C, Matlab, Shell, MySQL, HTML

BASICS Linux, Git, LaTeX, Docker

Teaching Experience

Spring 22 **CS 5984 Computational Social Science**, Teaching Assistant
 Fall 21 **CS 5984 Natural Language Processing**, Teaching Assistant
 Spring 21 **CS 2505 Computer Architecture I**, Teaching Assistant
 Fall 20 **CS 3214 Operating System**, Teaching Assistant